

An abstract digital background on the left side of the slide. It features glowing blue and yellow lines that resemble circuit traces or data paths, set against a dark blue background with soft, out-of-focus light spots in purple and blue.

Knowledge-Based Models: The Future of Software Fault Detection

How intelligent systems are transforming the way we find, diagnose, and recover from software failures — moving from reactive fixes to proactive resilience.

The Diagnostic Shift

Traditional Expert Systems

Built on rules-of-thumb and accumulated human experience. These systems are brittle — they lack robustness and completeness when faced with novel or complex failure scenarios.

Modern Model-Based Diagnosis (MBD)

Constructs **first-principles causal models** of system behavior, enabling mathematical reasoning about which components are misbehaving — without relying on prior human heuristics.

- ① MBD solves the **inverse problem**: given observed behavior, infer the faulty components.

Core Detection Methodologies

1

Model-Based Diagnosis (MBD)

Uses a formal system description and runtime observations to logically identify faulty components through consistency checking.

2

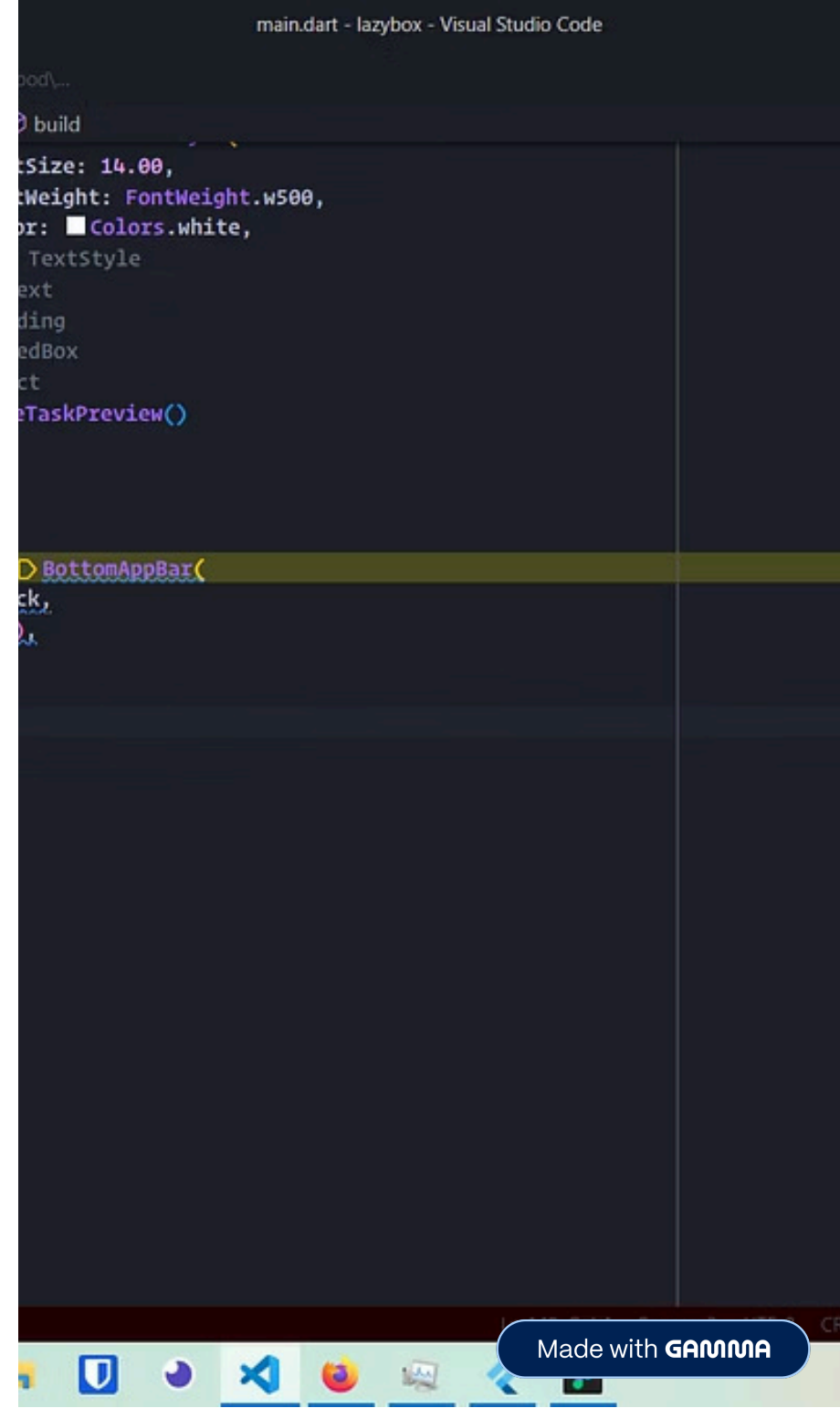
Spectrum-Based Fault Localization (SFL)

A statistical approach correlating program execution traces with test outcomes — isolating bugs **without expensive behavioral models**.

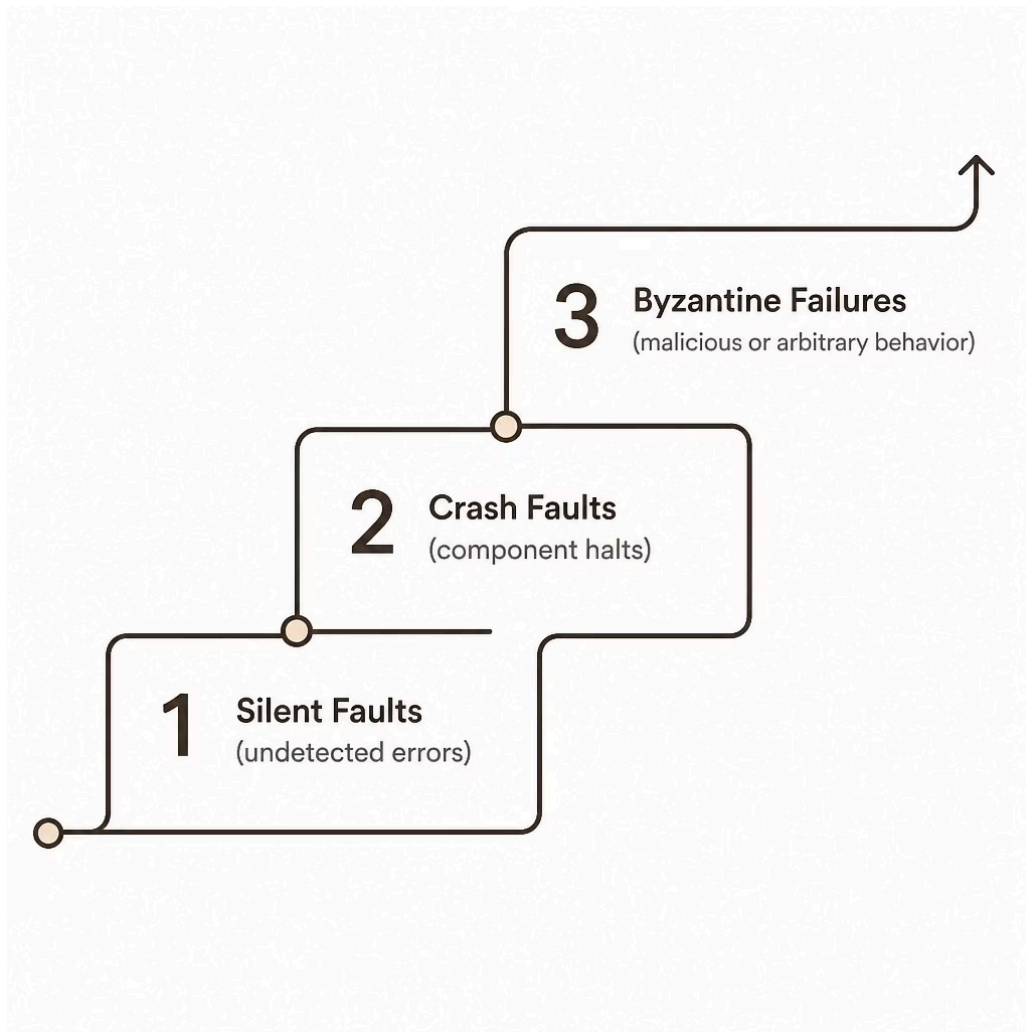
3

Control-Flow Checking (CFCSS)

Divides programs into basic blocks and computes runtime signatures to detect **illegal branch transitions** caused by faults.



Architecting Resilience



From Detection to Self-Healing

Modern architectures go beyond fault detection — they aim for **autonomous recovery**.

→ Fault Models

Categorize anomalies from silent crashes to Byzantine failures, defining the threat landscape.

→ Aspect-Oriented Modeling

Non-invasively weave self-healing logic into base system functionality without modifying core code.

→ Generative Programming

Automatically synthesize recovery routines tailored to detected fault patterns.